In [71]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [72]:
```python
import pandas as pd
```

In [73]:
```python
train = pd.read_csv(
    r"C:\Users\soura\Desktop\capston project report\Train.csv"
)

train.head()
```

Out[73]:

	Brand	Model_Info	Additional_Description	Locality	City	State	Price
0	1	name0 name234 64gb space grey	1yesr old mobile number 999two905two99 bill c...	878	8	2	15000
1	1	phone 7 name42 name453 new condition box acce...	101004800 1010065900 7000	1081	4	0	18800
2	1	name0 x 256gb leess used good condition	1010010000 seperate screen guard 3 back cover...	495	11	4	50000
3	1	name0 6s plus 64 gb space grey	without 1010020100 id 1010010300 colour 10100...	287	10	7	16500
4	1	phone 7 sealed pack brand new factory outet p...	101008700 10100000 xs max 64 gb made 10100850...	342	4	0	26499

In [74]:
```python
import os

print(os.getcwd())
print(os.listdir())
```

```
C:\Users\soura
['-1.14-windows.xml', '.anaconda', '.android', '.antigravity', '.bash_histor
y', '.conda', '.continuum', '.eclipse', '.emulator_console_auth_token', '.ge
mini', '.gitignore', '.gradle', '.ipynb_checkpoints', '.ipython', '.jupyte
r', '.Ld9VirtualBox', '.matplotlib', '.p2', '.vscode', '.vscode-shared', '3D
Objects', 'anaconda_projects', 'AndroidStudioProjects', 'AppData', 'Applicat
ion Data', 'basics_dart', 'BIT279.tmp', 'BIT9DDC.tmp', 'class_example', 'cmd
line-tools', 'commandlinetools-win-11076708_latest.zip', 'const_vs_final',
'Contacts', 'Cookies', 'Desktop', 'Document from Sourav.docx', 'Documents',
'Downloads', 'eclipse', 'eclipse-workspace', 'exception_handing', 'Favorite
s', 'firstproject_dart', 'functions', 'inheretance', 'IntelGraphicsProfile
s', 'Lib', 'Links', 'Local Settings', 'mixinsample', 'Music', 'My Document
s', 'NetHood', 'new1', 'nixinsample', 'NTUSER.DAT', 'ntuser.dat.LOG1', 'ntus
er.dat.LOG2', 'NTUSER.DAT{29cbf1ff-8ab4-11ee-a684-cc57229b91c3}.TxR.0.regtra
ns-ms', 'NTUSER.DAT{29cbf1ff-8ab4-11ee-a684-cc57229b91c3}.TxR.1.regtrans-m
s', 'NTUSER.DAT{29cbf1ff-8ab4-11ee-a684-cc57229b91c3}.TxR.2.regtrans-ms', 'N
TUSER.DAT{29cbf1ff-8ab4-11ee-a684-cc57229b91c3}.TxR.blf', 'NTUSER.DAT{29cbf2
00-8ab4-11ee-a684-cc57229b91c3}.TM.blf', 'NTUSER.DAT{29cbf200-8ab4-11ee-a684
-cc57229b91c3}.TMContainer00000000000000000001.regtrans-ms', 'NTUSER.DAT{29c
bf200-8ab4-11ee-a684-cc57229b91c3}.TMContainer00000000000000000002.regtrans-
ms', 'NTUSER.DAT{a6bccb45-052d-11ef-9498-dca2661bb17c}.TxR.0.regtrans-ms',
'NTUSER.DAT{a6bccb45-052d-11ef-9498-dca2661bb17c}.TxR.1.regtrans-ms', 'NTUSE
R.DAT{a6bccb45-052d-11ef-9498-dca2661bb17c}.TxR.2.regtrans-ms', 'NTUSER.DAT
{a6bccb45-052d-11ef-9498-dca2661bb17c}.TxR.blf', 'NTUSER.DAT{a6bccb46-052d-1
1ef-9498-dca2661bb17c}.TM.blf', 'NTUSER.DAT{a6bccb46-052d-11ef-9498-dca2661b
b17c}.TMContainer00000000000000000001.regtrans-ms', 'NTUSER.DAT{a6bccb46-052
d-11ef-9498-dca2661bb17c}.TMContainer00000000000000000002.regtrans-ms', 'ntu
ser.ini', 'OneDrive', 'Pictures', 'price .ipynb', 'PrintHood', 'PycharmProje
cts', 'Recent', 'sample_project', 'Saved Games', 'Scripts', 'Searches', 'Sen
dTo', 'Start Menu', 'Templates', 'Untitled.ipynb', 'Videos']
```

In [75]:

```
import pandas as pd

train = pd.read_csv(r"C:\Users\soura\Desktop\capston project report\Train.cs
test = pd.read_csv(r"C:\Users\soura\Desktop\capston project report\Test.csv"

print(train.shape)
print(test.shape)

train.head()
```

```
(2326, 7)
(997, 6)
```

Out[75]:

	Brand	Model_Info	Additional_Description	Locality	City	State	Price
0	1	name0 name234 64gb space grey	1yesr old mobile number 999two905two99 bill c...	878	8	2	15000
1	1	phone 7 name42 name453 new condition box acce...	101004800 1010065900 7000	1081	4	0	18800
2	1	name0 x 256gb leess used good condition	1010010000 seperate screen guard 3 back cover...	495	11	4	50000
3	1	name0 6s plus 64 gb space grey	without 1010020100 id 1010010300 colour 10100...	287	10	7	16500
4	1	phone 7 sealed pack brand new factory outet p...	101008700 10100000 xs max 64 gb made 10100850...	342	4	0	26499

In [76]:

```
train.shape
```

Out[76]:

```
(2326, 7)
```

In [77]:

```
train.columns
```

Out[77]:

```
Index(['Brand', 'Model_Info', 'Additional_Description', 'Locality', 'City',
       'State', 'Price'],
      dtype='object')
```

In [78]:

```
train.head()
```

Out[78]:

	Brand	Model_Info	Additional_Description	Locality	City	State	Price
0	1	name0 name234 64gb space grey	1yesr old mobile number 999two905two99 bill c...	878	8	2	15000
1	1	phone 7 name42 name453 new condition box acce...	101004800 1010065900 7000	1081	4	0	18800
2	1	name0 x 256gb leess used good condition	1010010000 seperate screen guard 3 back cover...	495	11	4	50000
3	1	name0 6s plus 64 gb space grey	without 1010020100 id 1010010300 colour 10100...	287	10	7	16500
4	1	phone 7 sealed pack brand new factory outet p...	101008700 10100000 xs max 64 gb made 10100850...	342	4	0	26499

In [79]:

```
train.isnull().sum()
```

Out[79]:

```
Brand                     0
Model_Info                0
Additional_Description    0
Locality                  0
City                      0
State                     0
Price                     0
dtype: int64
```

In [80]:

```
train.dtypes
```

Out[80]:

```
Brand                      int64
Model_Info                object
Additional_Description    object
Locality                   int64
City                       int64
State                      int64
Price                      int64
dtype: object
```

In [81]:

```
train.describe()
```

Out[81]:

	Brand	Locality	City	State	Price
count	2326.000000	2326.000000	2326.000000	2326.000000	2326.000000
mean	1.047291	538.894239	7.294067	3.693465	25562.137145
std	0.396109	333.355186	5.408113	2.194072	21316.854497
min	0.000000	0.000000	0.000000	0.000000	399.000000
25%	1.000000	249.000000	2.000000	2.000000	12000.000000
50%	1.000000	534.000000	8.000000	4.000000	18945.000000
75%	1.000000	814.000000	11.000000	5.000000	30974.250000
max	3.000000	1191.000000	17.000000	8.000000	129998.000000

In [82]:

```
train.info()
```

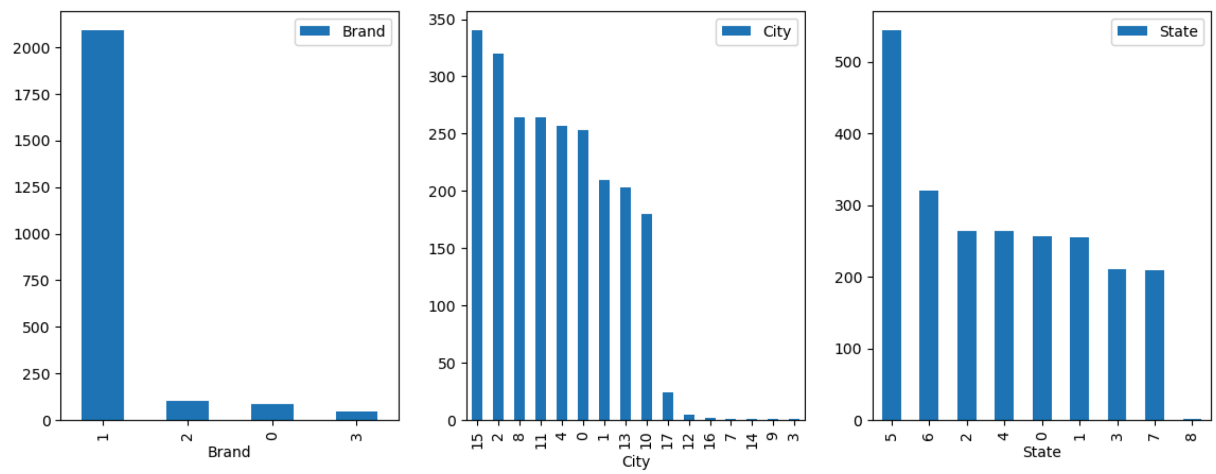
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2326 entries, 0 to 2325
Data columns (total 7 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Brand                   2326 non-null   int64
 1   Model_Info              2326 non-null   object
 2   Additional_Description  2326 non-null   object
 3   Locality                2326 non-null   int64
 4   City                    2326 non-null   int64
 5   State                   2326 non-null   int64
 6   Price                   2326 non-null   int64
dtypes: int64(5), object(2)
memory usage: 127.3+ KB
```

In [83]:

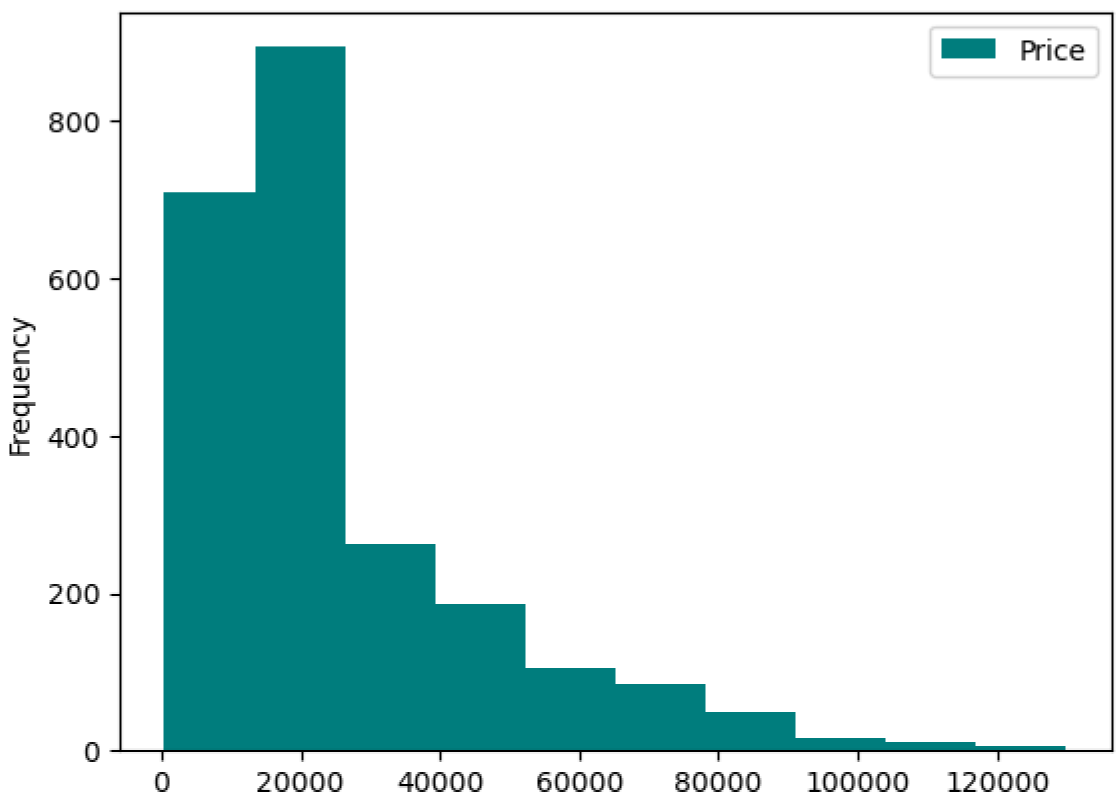
```
import matplotlib.pyplot as plt
```

In [84]:

```
import matplotlib.pyplot as plt

plt.figure(figsize=(14,5))

plt.subplot(1,3,1)
train.Brand.value_counts().plot(kind='bar', label='Brand')
plt.legend()

plt.subplot(1,3,2)
train.City.value_counts().plot(kind='bar', label='City')
plt.legend()

plt.subplot(1,3,3)
train.State.value_counts().plot(kind='bar', label='State')
plt.legend()

plt.show()
```



In [85]:
```python
train.Price.plot(kind='hist', color=['teal'], label="Price")
plt.legend()
plt.show()
```



In [86]:
```python
def device_memory_category(x):
    if (x.find("16gb") != -1) or (x.find("16 gb") != -1):
        return 1
    elif(x.find("32gb") != -1) or (x.find("32 gb") != -1):
        return 2
    elif(x.find("64gb") != -1) or (x.find("64 gb") != -1):
        return 3
    elif(x.find("128gb") != -1) or (x.find("128 gb") != -1):
        return 4
    elif(x.find("256gb") != -1) or (x.find("256 gb") != -1):
        return 1
    else:
```

```
        return 0

def if_iphopne_or_ipad(x):
    if (x.find("iphone") != -1) or (x.find("ipad") != -1):
        return 1
    else:
        return 0

def device_condition(x):
    if (x.find("good") != -1) or (x.find("great") != -1) or (x.find("excelle
    or (x.find("mint") != -1):
        return 1
    else:
        return 0

def under_warranty(x):
    if (x.find("billbox") != -1) or (x.find("warranty") != -1) or (x.find("b
    or (x.find("bill box") != -1):
        return 1
    else:
        return 0
```

In [87]:
```
train["device_memory"] = train["Model_Info"].apply(lambda x: device_memory_c
train["phone_status"] = train["Model_Info"].apply(lambda x: if_iphopne_or_ip
train["device_condition"] = train["Model_Info"].apply(lambda x: device_condi
train["warranty_status"] = train["Additional_Description"].apply(lambda x: u
```

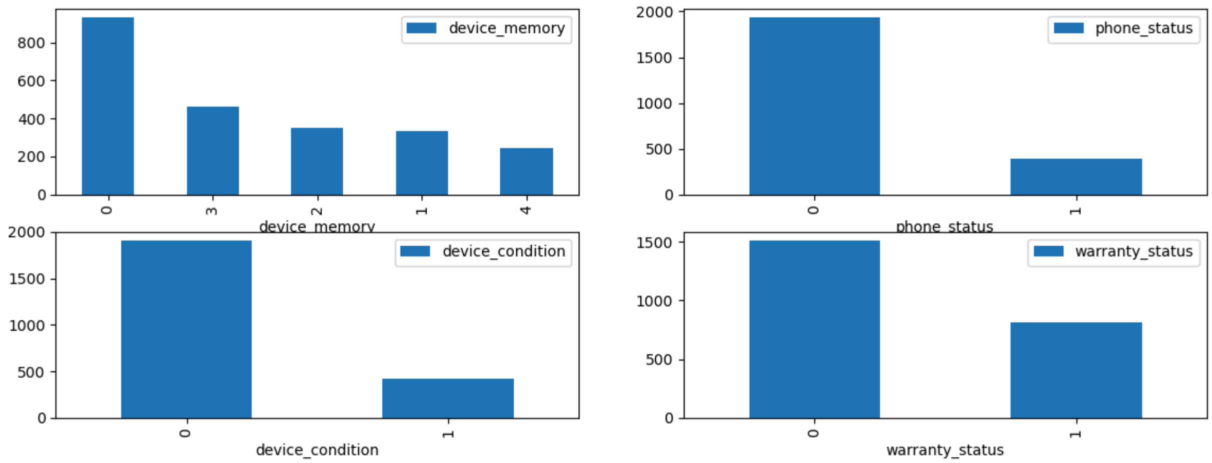
In [88]:
```
train.head()
```

Out[88]:

	Brand	Model_Info	Additional_Description	Locality	City	State	Price	dev
0	1	name0 name234 64gb space grey	1yesr old mobile number 999two905two99 bill c...	878	8	2	15000	
1	1	phone 7 name42 name453 new condition box acce...	101004800 1010065900 7000	1081	4	0	18800	
2	1	name0 x 256gb leess used good condition	1010010000 seperate screen guard 3 back cover...	495	11	4	50000	
3	1	name0 6s plus 64 gb space grey	without 1010020100 id 1010010300 colour 10100...	287	10	7	16500	
4	1	phone 7 sealed pack brand new factory outet p...	101008700 10100000 xs max 64 gb made 10100850...	342	4	0	26499	

In [89]:
```python
plt.figure(figsize=(14,5))

plt.subplot(2,2,1)
train.device_memory.value_counts().plot(kind='bar', label = 'device_memory')
plt.legend()

plt.subplot(2,2,2)
train.phone_status.value_counts().plot(kind='bar', label = 'phone_status')
plt.legend()

plt.subplot(2,2,3)
train.device_condition.value_counts().plot(kind='bar', label = 'device_condi
plt.legend()

plt.subplot(2,2,4)
train.warranty_status.value_counts().plot(kind='bar', label = 'warranty_stat
plt.legend()
```

Out[89]: <matplotlib.legend.Legend at 0x1b4bf4efd90>



In [90]:
```python
train.columns
```

Out[90]:
```
Index(['Brand', 'Model_Info', 'Additional_Description', 'Locality', 'City',
       'State', 'Price', 'device_memory', 'phone_status', 'device_conditio
n',
       'warranty_status'],
      dtype='object')
```

In [91]:
```python
train = pd.get_dummies(data=train, columns=['City','State','device_memory'],
                       prefix=['City', 'State', 'Device_memory'], drop_first=True)
```

In [92]:
```python
train['City_5'] = 0
train['City_6'] = 0
train['City_18'] = 0
train['City_19'] = 0
```

In [93]:
```python
train.shape
```

Out[93]: (2326, 39)

In [94]:
```
train.columns
```

Out[94]:
```
Index(['Brand', 'Model_Info', 'Additional_Description', 'Locality', 'Pric
e',
       'phone_status', 'device_condition', 'warranty_status', 'City_1',
       'City_2', 'City_3', 'City_4', 'City_7', 'City_8', 'City_9', 'City_1
0',
       'City_11', 'City_12', 'City_13', 'City_14', 'City_15', 'City_16',
       'City_17', 'State_1', 'State_2', 'State_3', 'State_4', 'State_5',
       'State_6', 'State_7', 'State_8', 'Device_memory_1', 'Device_memory_
2',
       'Device_memory_3', 'Device_memory_4', 'City_5', 'City_6', 'City_18',
       'City_19'],
      dtype='object')
```

In [95]:
```
train = train[['Brand', 'Model_Info', 'Additional_Description', 'Locality',
       'phone_status', 'device_condition', 'warranty_status', 'City_1',
       'City_2', 'City_3', 'City_4', 'City_5', 'City_6', 'City_7', 'City_8',
       'City_11', 'City_12', 'City_13', 'City_14', 'City_15', 'City_16',
       'City_17', 'City_18', 'City_19', 'State_1', 'State_2', 'State_3', 'St
       'State_6', 'State_7', 'State_8', 'Device_memory_1', 'Device_memory_2'
       'Device_memory_3', 'Device_memory_4']]
```

In [96]:
```
train.reset_index(drop=True, inplace=True)
```

In [97]:
```
x = train.drop(['Price', 'Model_Info', 'Additional_Description'], axis=1)
y = train['Price']
```

In [ ]:
```
x_copy = x.copy()
```

In [ ]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size = 0.2, r
```

In [ ]:
```
from sklearn.linear_model import ElasticNet, Lasso,  BayesianRidge, LassoLar
from sklearn.ensemble import RandomForestRegressor,  GradientBoostingRegress
from sklearn.kernel_ridge import KernelRidge
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import RobustScaler
from sklearn.base import BaseEstimator, TransformerMixin, RegressorMixin, cl
from sklearn.model_selection import KFold, cross_val_score, train_test_split
from sklearn.metrics import mean_squared_error
```

In [101…
```
import xgboost as xgb
```

In [102…
```
import lightgbm as lgb
```

In [103…
```
from sklearn.pipeline import Pipeline
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import roc_curve, auc, classification_report, confusion
from sklearn.metrics import classification_report
```

In [104…
```
n_folds = 5
```

```
def rmsle_cv(model):
    kf = KFold(n_folds, shuffle=True, random_state=42).get_n_splits(train.va
    rmse= np.sqrt(-cross_val_score(model, X_train.values, y_train, scoring="
    return(rmse)
```

In [105…

```
lasso = make_pipeline(RobustScaler(), Lasso(alpha =0.0005, random_state=1))
```

In [106…

```
ENet = make_pipeline(RobustScaler(), ElasticNet(alpha=0.0005, l1_ratio=.9, r
```

In [107…

```
KRR = KernelRidge(alpha=0.6, kernel='polynomial', degree=2, coef0=2.5)
GBoost = GradientBoostingRegressor(n_estimators=3000, learning_rate=0.05,
                                   max_depth=4, max_features='sqrt',
                                   min_samples_leaf=15, min_samples_split=10
                                   loss='huber', random_state =5)
```

In [108…

```
model_xgb = xgb.XGBRegressor(colsample_bytree=0.4603, gamma=0.0468,
                             learning_rate=0.05, max_depth=3,
                             min_child_weight=1.7817, n_estimators=2200,
                             reg_alpha=0.4640, reg_lambda=0.8571,
                             subsample=0.5213, silent=1,
                             random_state =7, nthread = -1)
```

In [109…

```
model_lgb = lgb.LGBMRegressor(objective='regression',num_leaves=5,
                              learning_rate=0.05, n_estimators=720,
                              max_bin = 55, bagging_fraction = 0.8,
                              bagging_freq = 5, feature_fraction = 0.2319,
                              feature_fraction_seed=9, bagging_seed=9,
                              min_data_in_leaf =6, min_sum_hessian_in_leaf =
```

In [110…

```
score = rmsle_cv(lasso)
print("\nLasso score: {:.4f} ({:.4f})\n".format(score.mean(), score.std()))

score = rmsle_cv(ENet)
print("ElasticNet score: {:.4f} ({:.4f})\n".format(score.mean(), score.std()

score = rmsle_cv(KRR)
print("Kernel Ridge score: {:.4f} ({:.4f})\n".format(score.mean(), score.std

score = rmsle_cv(GBoost)
print("Gradient Boosting score: {:.4f} ({:.4f})\n".format(score.mean(), scor
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 1.001e+11, tolerance: 6.942
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 4.813e+10, tolerance: 7.098
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 1.079e+11, tolerance: 7.097
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 9.212e+10, tolerance: 6.926
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 1.149e+11, tolerance: 6.923
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 2.930e+11, tolerance: 6.942
e+07
  model = cd_fast.enet_coordinate_descent(
Lasso score: 20163.6531 (730.4955)
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 2.967e+11, tolerance: 7.098
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 2.965e+11, tolerance: 7.097
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 2.829e+11, tolerance: 6.926
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 2.917e+11, tolerance: 6.923
e+07
  model = cd_fast.enet_coordinate_descent(
```

```
ElasticNet score: 20163.0262 (727.2152)

Kernel Ridge score: 20003.7316 (718.0619)

Gradient Boosting score: 20072.8876 (716.8621)
```

In [111…

```
score = rmsle_cv(model_xgb)
print("Xgboost score: {:.4f} ({:.4f})\n".format(score.mean(), score.std()))


score = rmsle_cv(model_lgb)
print("LGBM score: {:.4f} ({:.4f})\n" .format(score.mean(), score.std()))
```

```
C:\Users\soura\AppData\Roaming\Python\Python313\site-packages\xgboost\traini
ng.py:200: UserWarning: [23:04:28] WARNING: C:\actions-runner\_work\xgboost
\xgboost\src\learner.cc:782:
Parameters: { "silent" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
C:\Users\soura\AppData\Roaming\Python\Python313\site-packages\xgboost\traini
ng.py:200: UserWarning: [23:04:30] WARNING: C:\actions-runner\_work\xgboost
\xgboost\src\learner.cc:782:
Parameters: { "silent" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
C:\Users\soura\AppData\Roaming\Python\Python313\site-packages\xgboost\traini
ng.py:200: UserWarning: [23:04:32] WARNING: C:\actions-runner\_work\xgboost
\xgboost\src\learner.cc:782:
Parameters: { "silent" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
C:\Users\soura\AppData\Roaming\Python\Python313\site-packages\xgboost\traini
ng.py:200: UserWarning: [23:04:34] WARNING: C:\actions-runner\_work\xgboost
\xgboost\src\learner.cc:782:
Parameters: { "silent" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
C:\Users\soura\AppData\Roaming\Python\Python313\site-packages\xgboost\traini
ng.py:200: UserWarning: [23:04:36] WARNING: C:\actions-runner\_work\xgboost
\xgboost\src\learner.cc:782:
Parameters: { "silent" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
```

```
Xgboost score: 20335.8908 (572.7295)

[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000238 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 105
[LightGBM] [Info] Number of data points in the train set: 1488, number of us
ed features: 25
[LightGBM] [Info] Start training from score 25812.102151
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749:
UserWarning: X does not have valid feature names, but LGBMRegressor was fitt
ed with feature names
  warnings.warn(
```

```
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000166 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 105
[LightGBM] [Info] Number of data points in the train set: 1488, number of us
ed features: 25
[LightGBM] [Info] Start training from score 26098.547715
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749:
UserWarning: X does not have valid feature names, but LGBMRegressor was fitt
ed with feature names
  warnings.warn(
```

```
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of te
sting was 0.000281 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 105
[LightGBM] [Info] Number of data points in the train set: 1488, number of us
ed features: 25
[LightGBM] [Info] Start training from score 26025.146505
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749:
UserWarning: X does not have valid feature names, but LGBMRegressor was fitt
ed with feature names
  warnings.warn(
```

```
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000197 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 105
[LightGBM] [Info] Number of data points in the train set: 1488, number of us
ed features: 25
[LightGBM] [Info] Start training from score 26012.590726
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749:
UserWarning: X does not have valid feature names, but LGBMRegressor was fitt
ed with feature names
  warnings.warn(
```

```
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 will be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.001 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be ignored. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignored. Current value: bagging_freq=5
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 will be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.001 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be ignored. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignored. Current value: bagging_freq=5
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 will be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.001 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be ignored. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignored. Current value: bagging_freq=5
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of testing was 0.000306 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 105
[LightGBM] [Info] Number of data points in the train set: 1488, number of used features: 25
[LightGBM] [Info] Start training from score 25935.825269
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 will be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.001 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be ignored. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignored. Current value: bagging_freq=5
LGBM score: 19367.1209 (584.2444)
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but LGBMRegressor was fitted with feature names
  warnings.warn(
```

In [114…

```
class AveragingModels(BaseEstimator, RegressorMixin, TransformerMixin):
```

```python
    def __init__(self, models):
        self.models = models

    def fit(self, X, y):
        self.models_ = [clone(x) for x in self.models]

        for model in self.models_:
            model.fit(X, y)

        return self

    def predict(self, X):
        predictions = np.column_stack([
            model.predict(X) for model in self.models_
        ])

        return np.mean(predictions, axis=1)
```

In [119…

```python
averaged_models = AveragingModels(models = (ENet, GBoost, KRR, lasso))

score = rmsle_cv(averaged_models)
print(" Averaged base models score: {:.4f} ({:.4f})\n".format(score.mean(),
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 2.930e+11, tolerance: 6.942
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 1.001e+11, tolerance: 6.942
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 2.967e+11, tolerance: 7.098
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 4.813e+10, tolerance: 7.098
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 2.965e+11, tolerance: 7.097
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 1.079e+11, tolerance: 7.097
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 2.829e+11, tolerance: 6.926
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 9.212e+10, tolerance: 6.926
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 2.917e+11, tolerance: 6.923
e+07
  model = cd_fast.enet_coordinate_descent(
```

```
Averaged base models score: 19638.3130 (727.1585)
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 1.149e+11, tolerance: 6.923
e+07
  model = cd_fast.enet_coordinate_descent(
```

In [121…

```
model = averaged_models.fit(X_train, y_train)
predictions = model.predict(X_test)
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 3.662e+11, tolerance: 8.747
e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 1.229e+11, tolerance: 8.747
e+07
  model = cd_fast.enet_coordinate_descent(
```

In [122…

```
mean_squared_error(predictions, y_test)
```

Out[122…

```
354506433.56866646
```

In [125…

```
model_full = averaged_models.fit(x, y)
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 4.498e+11, tolerance: 1.056
e+08
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_
descent.py:695: ConvergenceWarning: Objective did not converge. You might wa
nt to increase the number of iterations, check the scale of the features or
consider increasing regularisation. Duality gap: 1.629e+11, tolerance: 1.056
e+08
  model = cd_fast.enet_coordinate_descent(
```

In [126…

```
lightgbm_full = model_lgb.fit(x, y)
```

```
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 will be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.001 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be ignored. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignored. Current value: bagging_freq=5
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 will be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.001 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be ignored. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignored. Current value: bagging_freq=5
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of testing was 0.000239 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 105
[LightGBM] [Info] Number of data points in the train set: 2326, number of used features: 25
[LightGBM] [Info] Start training from score 25562.137145
```

In [127…

```
model_xgb_full = model_xgb.fit(x, y)
```

```
C:\Users\soura\AppData\Roaming\Python\Python313\site-packages\xgboost\training.py:200: UserWarning: [23:26:49] WARNING: C:\actions-runner\_work\xgboost\xgboost\src\learner.cc:782:
Parameters: { "silent" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
```

In [129…

```
test = pd.read_csv(r"C:\Users\soura\Desktop\capston project report\Test.csv"
test.shape
```

Out[129… (997, 6)

In [130…

```
test.head(5)
```

Out[130…

	Brand	Model_Info	Additional_Description	Locality	City	State
0	1	name0 55s66s66s778xxsxsmax etc	good condition 11months old single scratch we...	570	11	4
1	1	slightly used excellent condition name0 5 sale	101008700 1010030600 1010034300 10100192200 1...	762	8	2
2	1	name0 sx ios12 top letast model bill call	1010017300 delivery	60	13	5
3	1	name87 name0 x 64gb going lowest 41900	phone 1010023400 64 gb excellent condition sale	640	15	5
4	1	name0 5s proper condition one handedly used	full kit available 10100248300 condition 4gb ...	816	2	6

In [131…

```
test["device_memory"] = test["Model_Info"].apply(lambda x: device_memory_cat
test["phone_status"] = test["Model_Info"].apply(lambda x: if_iphopne_or_ipac
test["device_condition"] = test["Model_Info"].apply(lambda x: device_conditi
test["warranty_status"] = test["Additional_Description"].apply(lambda x: unc
```

In [132…

```
test = pd.get_dummies(data=test, columns=['City','State','device_memory'],
               prefix=['City', 'State', 'Device_memory'],
               drop_first=True)
```

In [133…

```
test.columns
```

Out[133…

```
Index(['Brand', 'Model_Info', 'Additional_Description', 'Locality',
       'phone_status', 'device_condition', 'warranty_status', 'City_1',
       'City_2', 'City_4', 'City_5', 'City_6', 'City_8', 'City_10', 'City_1
1',
       'City_12', 'City_13', 'City_15', 'City_17', 'City_18', 'City_19',
       'State_1', 'State_2', 'State_3', 'State_4', 'State_5', 'State_6',
       'State_7', 'Device_memory_1', 'Device_memory_2', 'Device_memory_3',
       'Device_memory_4'],
      dtype='object')
```

In [134…

```
test['City_3'] = 0
test['City_7'] = 0
test['City_9'] = 0
test['City_14'] = 0
test['City_16'] = 0
test['State_8'] = 0
```

In [135…

```
test_for_prediction = test[['Brand', 'Locality',
       'phone_status', 'device_condition', 'warranty_status', 'City_1',
       'City_2', 'City_3', 'City_4', 'City_5', 'City_6', 'City_7', 'City_8',
       'City_11', 'City_12', 'City_13', 'City_14', 'City_15', 'City_16',
       'City_17', 'City_18', 'City_19', 'State_1', 'State_2', 'State_3', 'St
       'State_6', 'State_7', 'State_8', 'Device_memory_1', 'Device_memory_2'
       'Device_memory_3', 'Device_memory_4']]
```

In [136…

```
test_for_prediction.head()
```

Out[136…

	Brand	Locality	phone_status	device_condition	warranty_status	City_1	C
0	1	570	0	0	0	False	
1	1	762	0	1	0	False	
2	1	60	0	0	0	False	
3	1	640	0	0	0	False	
4	1	816	0	0	0	False	

5 rows × 36 columns

In [137…

```
def predict_file(model, model_instance, test_data):
    file_name = "Final_output_prediction_from_" + model + ".xlsx"
    predictions  = model_instance.predict(test_data)
    df_prediction_var = pd.DataFrame(predictions, columns=["Price"])
    df_prediction_var.to_excel(file_name)
    print("{} created.".format(file_name))
```

In [138…

```
predict_file("stacked_model", model_full, test_for_prediction)
```

```
Final_output_prediction_from_stacked_model.xlsx created.
```

In [139…

```
predict_file("lightgbm_model", lightgbm_full, test_for_prediction)
```

```
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
Final_output_prediction_from_lightgbm_model.xlsx created.
```

In [140…

```
predict_file("xgboost_model", model_xgb_full, test_for_prediction)
```

```
Final_output_prediction_from_xgboost_model.xlsx created.
```

In [141…

```
stacked_pred = model_full.predict(test_for_prediction.values)

xgb_pred = model_xgb_full.predict(test_for_prediction)

lgb_pred = lightgbm_full.predict(test_for_prediction.values)

ensemble = stacked_pred*0.70 + xgb_pred*0.15 + lgb_pred*0.15
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749:
UserWarning: X does not have valid feature names, but RobustScaler was fitte
d with feature names
  warnings.warn(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749:
UserWarning: X does not have valid feature names, but GradientBoostingRegres
sor was fitted with feature names
  warnings.warn(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749:
UserWarning: X does not have valid feature names, but KernelRidge was fitted
with feature names
  warnings.warn(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749:
UserWarning: X does not have valid feature names, but RobustScaler was fitte
d with feature names
  warnings.warn(
[LightGBM] [Warning] min_data_in_leaf is set=6, min_child_samples=20 will be
ignored. Current value: min_data_in_leaf=6
[LightGBM] [Warning] feature_fraction is set=0.2319, colsample_bytree=1.0 wi
ll be ignored. Current value: feature_fraction=0.2319
[LightGBM] [Warning] min_sum_hessian_in_leaf is set=11, min_child_weight=0.0
01 will be ignored. Current value: min_sum_hessian_in_leaf=11
[LightGBM] [Warning] bagging_fraction is set=0.8, subsample=1.0 will be igno
red. Current value: bagging_fraction=0.8
[LightGBM] [Warning] bagging_freq is set=5, subsample_freq=0 will be ignore
d. Current value: bagging_freq=5
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749:
UserWarning: X does not have valid feature names, but LGBMRegressor was fitt
ed with feature names
  warnings.warn(
```

In [142…

```
ensemble_sub = pd.DataFrame()
ensemble_sub['Price'] = ensemble
ensemble_sub.to_excel('ensemble_submission.xlsx',index=False)
```

In [143…

```
print(type(lasso))
print(type(ENet))
print(type(KRR))
print(type(GBoost))
```

```
<class 'sklearn.pipeline.Pipeline'>
<class 'sklearn.pipeline.Pipeline'>
<class 'sklearn.kernel_ridge.KernelRidge'>
<class 'sklearn.ensemble._gb.GradientBoostingRegressor'>
```

In [144…

```
averaged_models = AveragingModels(models=(ENet, GBoost, KRR, lasso))
```

In [145…

```
model = averaged_models.fit(X_train, y_train)
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 3.662e+11, tolerance: 8.747e+07
  model = cd_fast.enet_coordinate_descent(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model\_coordinate_descent.py:695: ConvergenceWarning: Objective did not converge. You might want to increase the number of iterations, check the scale of the features or consider increasing regularisation. Duality gap: 1.229e+11, tolerance: 8.747e+07
  model = cd_fast.enet_coordinate_descent(
```

In [146…

```python
predictions = model.predict(X_test)
```

In [147…

```python
print(len(predictions))
```

```
466
```

In [148…

```python
from sklearn.metrics import mean_squared_error

mse = mean_squared_error(y_test, predictions)
print(mse)
```

```
354506433.56866646
```

In [149…

```python
test = pd.read_csv(r"C:\Users\soura\Desktop\capston project report\Test.csv"
```

In [150…

```python
test.shape
```

Out[150…

```
(997, 6)
```

In [ ]:

```
ValueError                                Traceback (most recent call last)
Cell In[152], line 1
----> 1 ensemble = model.predict(test)

Cell In[114], line 16, in AveragingModels.predict(self, X)
     14 def predict(self, X):
     15     predictions = np.column_stack([
---> 16         model.predict(X) for model in self.models_
     17     ])
     19     return np.mean(predictions, axis=1)

File C:\ProgramData\anaconda3\Lib\site-packages\sklearn\pipeline.py:788, in
    786 if not _routing_enabled():
    787     for _, name, transform in self._iter(with_final=False):
--> 788         Xt = transform.transform(Xt)
    789     return self.steps[-1][1].predict(Xt, **params)
    791 # metadata routing enabled

File C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\_set_output.py
    314 @wraps(f)
    315 def wrapped(self, X, *args, **kwargs):
--> 316     data_to_wrap = f(self, X, *args, **kwargs)
    317     if isinstance(data_to_wrap, tuple):
    318         # only wrap the first output for cross decomposition
```

```
    319         return_tuple = (
    320             _wrap_data_with_container(method, data_to_wrap[0], X, se
    321             *data_to_wrap[1:],
    322         )

File C:\ProgramData\anaconda3\Lib\site-packages\sklearn\preprocessing\_data.
   1687 """Center and scale the data.
   1688
   1689 Parameters
   (...)   1697     Transformed array.
   1698 """
   1699 check_is_fitted(self)
-> 1700 X = validate_data(
   1701     self,
   1702     X,
   1703     accept_sparse=("csr", "csc"),
   1704     copy=self.copy,
   1705     dtype=FLOAT_DTYPES,
   1706     force_writeable=True,
   1707     reset=False,
   1708     ensure_all_finite="allow-nan",
   1709 )
   1711 if sparse.issparse(X):
   1712     if self.with_scaling:

File C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:
   2845 def validate_data(
   2846     _estimator,
   2847     /,
   (...)   2853     **check_params,
   2854 ):
   2855     """Validate input data and set or check feature names and counts
   2856
   2857     This helper function should be used in an estimator that require
   (...)   2927         validated.
   2928     """
-> 2929     _check_feature_names(_estimator, X, reset=reset)
   2930     tags = get_tags(_estimator)
   2931     if y is None and tags.target_tags.required:

File C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:
   2784 if not missing_names and not unexpected_names:
   2785     message += "Feature names must be in the same order as they were
-> 2787 raise ValueError(message)

ValueError: The feature names should match those that were passed during fit
Feature names unseen at fit time:
- Additional_Description
- City
- Model_Info
- State
Feature names seen at fit time, yet now missing:
- City_1
- City_10
- City_11
- City_12
```

```
- City_13
- ...
```

In [153…

```
print(test.columns)
```

```
Index(['Brand', 'Model_Info', 'Additional_Description', 'Locality', 'City',
       'State'],
      dtype='object')
```

In [154…

```
print(x.columns)
```

```
Index(['Brand', 'Locality', 'phone_status', 'device_condition',
       'warranty_status', 'City_1', 'City_2', 'City_3', 'City_4', 'City_5',
       'City_6', 'City_7', 'City_8', 'City_9', 'City_10', 'City_11', 'City_1
2',
       'City_13', 'City_14', 'City_15', 'City_16', 'City_17', 'City_18',
       'City_19', 'State_1', 'State_2', 'State_3', 'State_4', 'State_5',
       'State_6', 'State_7', 'State_8', 'Device_memory_1', 'Device_memory_
2',
       'Device_memory_3', 'Device_memory_4'],
      dtype='object')
```

In [155…

```
test_new = test.copy()
```

In [156…

```
test_new['Model_Info'] = test_new['Model_Info'].str.lower()
test_new['Additional_Description'] = test_new['Additional_Description'].str.
```

In [157…

```
test_new['device_memory'] = test_new['Model_Info'].apply(device_memory_categ

test_new['phone_status'] = test_new['Model_Info'].apply(if_iphopne_or_ipad)

test_new['device_condition'] = test_new['Model_Info'].apply(device_conditior

test_new['warranty_status'] = test_new['Additional_Description'].apply(under
```

In [158…

```
test_new = pd.get_dummies(
    data=test_new,
    columns=['City', 'State', 'device_memory'],
    prefix=['City', 'State', 'Device_memory'],
    drop_first=True
)
```

In [160…

```
for col in x.columns:
    if col not in test_new.columns:
        test_new[col] = 0
```

In [161…

```
test_new = test_new[x.columns]
```

In [162…

```
print(test_new.shape)
print(test_new.columns.equals(x.columns))
```

```
(997, 36)
True
```

In [163…

```
ensemble = model.predict(test_new)
```

In [164…

```
ensemble_sub = pd.DataFrame()
ensemble_sub['Price'] = ensemble

ensemble_sub.to_excel(
    'ensemble_submission.xlsx',
    index=False
)

print("File created successfully")
```

```
File created successfully
```

In [165…

```
import os
print(os.path.exists('ensemble_submission.xlsx'))
```

```
True
```

In [166…

```
ensemble = model.predict(test_new)
```

In [167…

```
ensemble_sub = pd.DataFrame()
ensemble_sub['Price'] = ensemble
```

In [168…

```
ensemble_sub.to_excel('ensemble_submission.xlsx', index=False)
print("File saved successfully")
```

```
File saved successfully
```

In [169…

```
import os
print(os.path.exists('ensemble_submission.xlsx'))
```

```
True
```

In [ ]:

```

```